

Tutorial 1

Matrices

1. *In mathematics you have to take definitions very seriously. They give you **all** the information about a new concept. If you want to use the new concept, the definition is the only information you have (at least until you have proved a few theorems). Learning to work with definitions is a very important mathematical skill. This question is designed to help you develop this skill.*

Read the following definitions carefully:

The *transpose* of an $n \times n$ matrix A is the matrix A^T obtained from A by interchanging the rows and columns of A .

An $n \times n$ matrix $A = (a_{ij})$ is called:

upper triangular if $a_{ij} = 0$ for all $i > j$,

diagonal if $a_{ij} = 0$ for all $i \neq j$,

symmetric if $A^T = A$,

skew-symmetric if $A^T = -A$.

Now do the following questions quickly in your group. Try to get agreement on the answers.

- (a) Give an example of a 3×3
 (i) upper triangular matrix (ii) diagonal matrix (iii) skew-symmetric matrix.

Solution: Here are possible examples:

$$(i) \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 5 \end{pmatrix} \quad (ii) \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix} \quad (iii) \begin{pmatrix} 0 & 1 & 2 \\ -1 & 0 & 3 \\ -2 & -3 & 0 \end{pmatrix}.$$

Ask a tutor to check your examples if you are not sure. Note that (i) an upper triangular matrix must have all the entries below the diagonal 0; the other entries could be non-zero or zero (ii) a diagonal matrix must have all the entries not on the diagonal 0; the entries on the diagonal could be zero or non-zero (iii) a skew symmetric matrix must have 0 in every position on the diagonal.

- (b) How do you think a **lower triangular matrix** should be defined? Write out your definition carefully, and then give an example of a 3×3 lower triangular matrix.

Solution: A $n \times n$ matrix is lower triangular if $a_{ij} = 0$ for all $i < j$.

Example: $\begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 0 & 5 \end{pmatrix}$.

- (c) Complete: If a square matrix A and its transpose are both upper triangular, then A is a matrix.

Solution: diagonal matrix.

- (d) Complete: A symmetric lower triangular matrix is a matrix.

Solution: ...is a diagonal matrix.

- (e) For each of the following statements, say whether it is true or false. Give a reason for your answer.

- i. A square matrix cannot be both upper and lower triangular.

Solution: False. Any diagonal matrix is both upper and lower triangular.

- ii. If A is a square matrix, $A - A^T$ is a skew-symmetric matrix.

Solution: True. $(A - A^T)^T = A^T - (A^T)^T = A^T - A = -(A - A^T)$. Since by definition a square matrix B is skew symmetric if and only if $B^T = -B$, this shows that $A - A^T$ is skew symmetric.

- iii. If A is an upper triangular matrix, then $a_{ij} \neq 0$ for all $i \leq j$.

Solution: False. (The definition only requires the elements below the diagonal to be zero; the elements on or above the diagonal could be zero or non-zero. In fact, for instance the zero matrix is upper triangular!)

- iv. If A is a diagonal matrix, then $a_{ii} \neq 0$ for every i .

Solution: False. (For instance, the zero matrix is a diagonal matrix!)

- v. If A is a diagonal matrix, there is a non-zero vector $\mathbf{x}$ and a scalar λ such that $A\mathbf{x} = \lambda\mathbf{x}$.

Solution: True. Let $A = \begin{pmatrix} a_{11} & & \\ & \ddots & \\ & & a_{nn} \end{pmatrix}$ be an $n \times n$ diagonal matrix and $1 \leq i \leq n$. Take $\lambda = a_{ii}$ and let $\mathbf{x}$ be the vector with 1 (or any non-zero number) in the i -th entry and 0 everywhere else. If you find it difficult to see where this came from, write the matrix equation for a 3×3 matrix and then write the matrix equation as a system of three linear equations.

- vi. If A is an upper triangular matrix, there is a non-zero vector $\mathbf{x}$ and a scalar λ such that $A\mathbf{x} = \lambda\mathbf{x}$.

Solution: True. In this case take $\lambda = a_{11}$ and $\mathbf{x}$ to be the vector with 1 (or any non-zero number) in the first position and 0 everywhere else.

2. Decide which of the following matrix products are defined, and where possible find the product.

(a) $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 1 & 1 \\ 1 & 0 \end{pmatrix}$

Solution: $\begin{pmatrix} 4 & 2 \\ 7 & 5 \end{pmatrix}$

(b) $\begin{pmatrix} -1 & 0 \\ 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$

Solution: $\begin{pmatrix} -1 & -2 & -3 \\ 5 & 7 & 9 \\ 1 & 2 & 3 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} -1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$

Solution: The product is undefined: The first matrix has 3 columns while the second matrix has only 2 rows.

(d) $\begin{pmatrix} x & y & z \end{pmatrix} \begin{pmatrix} a & d & e \\ d & b & f \\ e & f & c \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

Solution: The product of the three matrices is the following 1×1 matrix:

$$(ax^2 + by^2 + cz^2 + 2dxy + 2exz + 2fyz).$$

3. Find real numbers a, b, c, d, e satisfying the following equations. You should be able to do this just by looking; no writing required at all.

(a) $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} = a \begin{pmatrix} 1 \\ 4 \end{pmatrix} + b \begin{pmatrix} 2 \\ 5 \end{pmatrix} + c \begin{pmatrix} 3 \\ 6 \end{pmatrix}$

Solution: $a = -1$, $b = 1$, $c = 1$. (Note that these are the entries of the matrix $\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$).

(b) $\begin{pmatrix} -1 & 0 \\ 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 4 \end{pmatrix} = d \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} + e \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

Solution: $d = 1$, $e = 4$. (Note that these are the entries of the matrix $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$.)

4. Let $A = \begin{pmatrix} 2 & 3 \\ -1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 9 \\ -3 & k \end{pmatrix}$. What value(s) of k , if any, will make $AB = BA$?

Solution:

$$AB = \begin{pmatrix} 2 & 3 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 9 \\ -3 & k \end{pmatrix} = \begin{pmatrix} -7 & 18 + 3k \\ -4 & -9 + k \end{pmatrix}$$

while

$$BA = \begin{pmatrix} 1 & 9 \\ -3 & k \end{pmatrix} \begin{pmatrix} 2 & 3 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} -7 & 12 \\ -6 - k & -9 + k \end{pmatrix}$$

Then $AB = BA$ if and only if $18 + 3k = 12$ and $-4 = -6 - k$, which happens if and only if $k = -2$.

5. Let $A = \begin{pmatrix} 3 & -6 \\ -2 & 4 \end{pmatrix}$. Construct a 2×2 non-zero matrix B such that AB is the zero matrix.

Solution: Consider $B = \begin{pmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \end{pmatrix}$. Then $AB = 0$ is equivalent to

$$AB = \begin{pmatrix} 3 & -6 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \end{pmatrix} = \begin{pmatrix} 3b_{1,1} - 6b_{2,1} & 3b_{1,2} - 6b_{2,2} \\ -2b_{1,1} + 4b_{2,1} & -2b_{1,2} + 4b_{2,2} \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

That implies

$$b_{1,1} = 2b_{2,1}, \quad b_{1,2} = 2b_{2,2}$$

Then we can take for example

$$B = \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}.$$

(You can also see this more directly if you notice that column 2 of A equals -2 times column 1 of A .)

6. Prove that for any $m \times n$ matrix A and any $n \times r$ matrix B ,

$$(AB)^T = B^T A^T,$$

where T means transpose.

Solution: The (i, j) -entry of $(AB)^T$ is the (j, i) -entry of AB , which is

$$a_{j,1}b_{1,i} + \cdots + a_{j,n}b_{n,i}.$$

The entries in row i of B^T are $b_{1,i}, \dots, b_{n,i}$, because they come from column i of B . Likewise, the entries in column j of A^T are $a_{j,1}, \dots, a_{j,n}$, because they come from row j of A . The (i, j) -entry in $B^T A^T$ is therefore

$$a_{j,1}b_{1,i} + \cdots + a_{j,n}b_{n,i},$$

as above.

7. A square matrix P is called a *stochastic*, or *Markov matrix* if all its entries are non-negative and the entries in each column add up to 1. Prove that the product of two stochastic matrices is again a stochastic matrix.

Solution: As a hint, we give the proof for the 2×2 case; the general proof is similar. Try to write out the general proof for $n \times n$ matrices, using the definition of matrix multiplication.

$$\text{If } P = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} \text{ and } Q = \begin{pmatrix} q_{11} & q_{12} \\ q_{21} & q_{22} \end{pmatrix},$$

then

$$PQ = \begin{pmatrix} p_{11}q_{11} + p_{12}q_{21} & p_{11}q_{12} + p_{12}q_{22} \\ p_{21}q_{11} + p_{22}q_{21} & p_{21}q_{12} + p_{22}q_{22} \end{pmatrix}.$$

Since all the p_{ij} and q_{ij} are non-negative, the entries in PQ are also all non-negative (being sums of products of non-negative numbers). We also have

$$p_{11}q_{11} + p_{12}q_{21} + p_{21}q_{11} + p_{22}q_{21} = q_{11}(p_{11} + p_{21}) + q_{21}(p_{12} + p_{22}) = q_{11} + q_{21} = 1,$$

so the first column of PQ adds up to 1. The proof for the second column is similar.